

Worksheet — Chemical Reactions and Equations (Class 10 CBSE Science)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions. Show all balancing work and reasoning for the subjective questions.
Total time: 50 minutes.

Q1. The Law of Conservation of Mass requires: [1 mark]

- (A) mass of products = mass of reactants
- (B) mass of products > reactants
- (C) atoms can disappear
- (D) reactions release mass

Q2. In a redox reaction, oxidation and reduction always happen together. [1 mark]

- (A) True
- (B) False

Q3. $AB + CD \rightarrow AD + CB$ is a ____ reaction. [1 mark]

Q4. The slow eating-away of metals by air and moisture is called: [1 mark]

- (A) rancidity
- (B) corrosion
- (C) displacement
- (D) neutralisation

Q5. Identify: $2H_2O \rightarrow 2H_2O + O_2$. [2 marks]

- (A) combination
- (B) decomposition
- (C) displacement
- (D) double displacement

Q6. Balance: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$. Show atom counts on each side. [3 marks]

Q7. Define and give one example each of: combination, decomposition, double displacement. [3 marks]

Q8. In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, identify which substance is OXIDISED, which is REDUCED, and name the oxidising and reducing agents. [3 marks]

Q9. Why does food packaging often contain antioxidants? Give one example of an antioxidant and explain its role. [3 marks]

Q10. Balance the following four equations and classify each by reaction type. (a) $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$. (b) $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$. (c) $\text{AgNO}_3 + \text{KCl} \rightarrow \text{AgCl} + \text{KNO}_3$. (d) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$. [6 marks]

Q11. An iron gate left outside develops rust. (i) Why does this happen — chemically? (ii) Why is rusting more rapid in coastal areas than in the desert? (iii) Suggest THREE ways to slow or prevent rusting and explain WHY each works. [6 marks]

Answer Key

Worksheet — Chemical Reactions and Equations (Class 10 CBSE Science)

Q1. The Law of Conservation of Mass requires:

Answer: mass of products = mass of reactants

Atoms are conserved.

MCQ · EASY · 10-CBSE-SCI-CRE-03

Q2. In a redox reaction, oxidation and reduction always happen together.

Answer: True

By definition.

TRUE_FALSE · EASY · 10-CBSE-SCI-CRE-07

Q3. $AB + CD \rightarrow AD + CB$ is a _____ reaction.

Answer: double displacement

Two compounds exchange ions.

FILL_BLANK · EASY · 10-CBSE-SCI-CRE-04

Q4. The slow eating-away of metals by air and moisture is called:

Answer: corrosion

Definition.

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Q5. Identify: $2H_2O \rightarrow 2H_2 + O_2$.

Answer: decomposition

One reactant → multiple products.

MCQ · EASY · 10-CBSE-SCI-CRE-04

Q6. Balance: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$. Show atom counts on each side.

Answer: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. Atoms — Fe: 3=3; H: 8=8; O: 4=4 ✓.

Iron + steam → magnetite + hydrogen.

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Q7. Define and give one example each of: combination, decomposition, double displacement.

Answer: Combination: $A + B \rightarrow AB$. Example: $2Mg + O_2 \rightarrow 2MgO$. Decomposition: $AB \rightarrow A + B$. Example: $2H_2O \rightarrow 2H_2 + O_2$. Double displacement: $AB + CD \rightarrow AD + CB$. Example: $AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$.

Three classic types.

SHORT_ANSWER · EASY · 10-CBSE-SCI-CRE-04

Q8. In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, identify which substance is OXIDISED, which is REDUCED, and name the oxidising and reducing agents.

Answer: H_2 is OXIDISED (gains oxygen $\rightarrow$ becomes H_2O). CuO is REDUCED (loses oxygen $\rightarrow$ becomes Cu). The oxidising agent is CuO (provides oxygen). The reducing agent is H_2 (takes oxygen).

Standard redox identification.

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Q9. Why does food packaging often contain antioxidants? Give one example of an antioxidant and explain its role.

Answer: Antioxidants prevent the oxidation of fats and oils in food, which would otherwise cause RANCIDITY (bad smell and taste). Example: BHA (butylated hydroxyanisole) or BHT. Role: they react with oxygen / free radicals before the fats do, so the food stays fresh longer.

Real-world food chemistry.

SHORT_ANSWER · MEDIUM · 10-CBSE-SCI-CRE-08

Q10. Balance the following four equations and classify each by reaction type. (a) $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$. (b) $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$. (c) $\text{AgNO}_3 + \text{KCl} \rightarrow \text{AgCl} + \text{KNO}_3$. (d) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$.

Answer: (a) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$. Type: DISPLACEMENT (Na displaces H from water). (b) $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$. Type: DECOMPOSITION (already balanced). (c) $\text{AgNO}_3 + \text{KCl} \rightarrow \text{AgCl} + \text{KNO}_3$. Type: DOUBLE DISPLACEMENT (already balanced). (d) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$. Type: COMBINATION (already balanced).

Balance + classify four reactions.

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Q11. An iron gate left outside develops rust. (i) Why does this happen — chemically? (ii) Why is rusting more rapid in coastal areas than in the desert? (iii) Suggest THREE ways to slow or prevent rusting and explain WHY each works.

Answer: (i) Iron reacts slowly with oxygen and moisture from the air to form HYDRATED IRON(III) OXIDE ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$), commonly called rust. (ii) Coastal areas have HIGH humidity AND salt particles in the air, which act as electrolytes that accelerate the rusting reaction. Deserts are dry, so rusting is slow. (iii) Three prevention methods: (1) PAINTING — physical barrier between iron and air. (2) GALVANISING (zinc coating) — zinc oxidises preferentially (sacrificial protection). (3) ALLOYING with chromium / nickel to make stainless steel — Cr forms a passive oxide layer that prevents further corrosion.

Rust + prevention with reasoning.

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